



Leveraging LLMs for action item identification in Urdu meetings: Dataset creation and comparative analysis

Bareera Sadia ^a , Farah Adeeba ^{b,*}, Sana Shams ^a, Sarmad Hussain ^a

^a Center for Language Engineering (CLE), KICS, University of Engineering and Technology (UET), Lahore, 54890, Pakistan

^b Department of Computer Science, New Campus, University of Engineering and Technology (UET), Lahore, 54890, Pakistan

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ABSTRACT

In response to the increasing number of online meetings, automation of action items identification in online Urdu meetings, has become crucial. To serve this purpose, this research presents the first ever dataset and guidelines for annotating action items in code-mixed Urdu-English language. Collected dataset comprises of 240 recorded meetings, 600 fabricated action items, and 250 real meeting action items, totaling 2948 action items. We evaluated the efficiency and accuracy of various deep learning and machine learning models through a comparative analysis on a balanced dataset being discussed in Section 4.2. Additionally, three Large Language Models (LLMs) BLOOMZ, LLaMA, and GPT-3.5 were tested using zero-shot and few-shot configurations. BLOOMZ and LLaMA were specifically fine-tuned to enhance their performance in recognizing Urdu meeting action items. The fine-tuned model, ur_BLOOMZ-1b1, achieved the highest average F1 score of 0.94, surpassing all other traditional models. This study lays a solid foundation for future research in multilingual environments and advances our understanding of action item identification in Urdu meetings.

1. Introduction

In today's collaborative and information-sharing environment, online meetings have become more and more frequent as technology has evolved and the global workforce has shifted to remote work. In the context of online meetings, identification of action items presents significant challenges for a number of reasons. First of all, manual efforts to identify particular tasks and responsibilities may be overloaded by the enormous amount of information shared during these online conversations. It is challenging to precisely identify and record action items in real-time not only due to the abundance of data but also due to the conversational style of meetings. Additionally, language characteristics and cultural factors further complicate the process, particularly in environments with multilingual languages such as meetings in Urdu code-mixed language. Meeting transcripts generated with Automatic Speech Recognition (ASR) technologies may contain mistakes and inaccuracies that need to be carefully verified and corrected. Moreover, action items often emerge naturally from spontaneous conversations, making them challenging to capture within the flow of discussion. Therefore, it takes a lot of time and effort to identify and monitor action items manually, with a possibility of overlooked tasks and missed deadlines. This emphasizes the need for automated solutions to improve both accuracy and efficiency in this process.

Action items (Purver, Ehlen, & Niekrasz, 2006) are decisions made during a meeting that require a substantial amount of discussion and a commitment from attendees. Action items are usually one or more statements made during a meeting that

* Corresponding author.

E-mail addresses: bareera.sadia@kics.edu.pk (B. Sadia), farah.adeeba@uet.edu.pk (F. Adeeba), sana.shams@kics.edu.pk (S. Shams), sarmad.hussain@kics.edu.pk (S. Hussain).

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summarize decisions that need to be paid attention or tasks that need to be completed after meeting. This allows participants to quickly understand the important tasks during the meeting without having to go through unnecessary information. Considerable progress has been made in the area of action items detection in meetings in the last several years. The use of automated methods for locating and organizing action items has increased with the development of sophisticated Natural Language Processing (NLP) tools. Researchers have used different machine learning and deep learning techniques to create complex models and algorithms for action item recognition. With the use of these technologies, actionable tasks can be more accurately and efficiently extracted from meeting transcripts, reducing the need for human involvement.

1.1. Motivation

Considering the rapid and unstructured nature of meetings, manually identifying and documenting action items is a difficult and time-consuming process. It is difficult to record important decisions and responsibilities in the midst of the discussion in real-time, and this process is made even more difficult in bilingual contexts, where conversational flow and cultural variations might make taking precise notes more difficult. While automated action items identification systems have demonstrated encouraging outcomes in English-speaking contexts, they are not as effective when used in languages with different linguistic patterns, like Urdu, especially in code-mixed environments. Urdu, the national language of our country, Pakistan, is frequently used in professional settings, often mixed with English, resulting in a complicated linguistic mix that poses further difficulties for NLP models that were predominantly trained on English data. Since there are no publicly available corpora for Urdu meetings with annotated action items, the lack of specialized resources makes it much more difficult to develop efficient action item identification system in areas like Pakistan. This gap emphasizes the need for a specific Urdu dataset that documents the code-mixing, conversational style, and cultural background common in these meetings.

Our research follows this path, with a particular emphasis on the detection of action items (Asthana, Hilleli, He, & Halfaker, 2023) in Urdu meetings. Unlike prior efforts that encounter the limitations of available corpora, a groundbreaking approach is used by collecting a comprehensive dataset that is specific to Urdu. This dataset combines a variety of sources, such as recorded meeting transcripts, fabricated data for action items by designing some scenarios, and action items taken from real meetings. Besides corpus development, action item identification, which is commonly formulated as a sentence-level binary classification problem, has also been tackled using a variety of techniques, ranging from classical machine learning and deep learning techniques to more modern approaches based on LLMs (Naveed et al., 2024).

1.2. Research objectives and contributions

The main objectives of this research are listed below:

- To develop guidelines for the annotation of action items.
- To develop a comprehensive corpus for action items by annotating meeting transcripts using developed guidelines.
- To compare the performance of different machine learning, deep learning models in the context of action items identification.
- To perform zero-shot and few-shot experiments on LLMs and to do fine-tuning of BLOOMZ and LLaMA for the identification of action items.

To fulfill these objectives, following are the key contributions of this research:

- **Development of Annotation Guidelines:** Thorough guidelines have been developed for the annotation of action items in Urdu meetings. With the goal of improving consistency and accuracy in the annotation process, these guidelines define clear criteria for recognizing and classifying action items.
- **Action Items Corpus Development:** Three primary methods have been used to generate the action items corpus. First, 240 transcripts of recorded meetings have been carefully annotated using the developed guidelines (as previously mentioned). Second, a fabricated data for only action items have been created by assuming different scenarios. Thirdly, actual meetings are attended and action items are exclusively noted down. Together, these three approaches helped to produce an extensive dataset for the identification and analysis of action items. This dataset has been made available in csv format on github.¹
- **Comparison of Models in the Context of Action Items Identification:** Various deep learning and machine learning models have been used, and their effectiveness in identifying action items has been assessed. The accuracy and efficiency of these algorithms in precisely identifying and classifying action items in meeting transcripts have been evaluated through a comparative analysis.
- **Evaluating and Fine-tuning LLMs:** Three of the LLMs have been used for zero-shot and few-shot experiments. Within these, BLOOMZ and LLaMA have been fine-tuned as well for more efficiently identifying action items in Urdu meetings.

The structure of the rest of the paper is as follows: Section 2 provides a detailed overview of the extensive array of related work including the state of the art work in the context of action items detection. Section 3 presents the complete development of action items corpus along with the developed guidelines. Sections 4 and 5 is based on the experimentation of different models and their results respectively. Finally, Sections 6–8 provide the conclusion, limitations and the future work of the research, respectively.

¹ https://github.com/BareeraSadia/Action_Items_Detection/tree/main/Dataset

2. Related work

Within the dynamic field of NLP, action item identification within virtual meetings is becoming a more and more important subject for researchers to work on. The growing use of virtual communication emphasizes the critical need for effective action item identification in the context of an information overload. Action item detection has been the subject of several studies, which is an indication of the increasing importance of this field of study. The purpose of this literature review is to shed light on the important contributions made by researchers and the state of the art in this rapidly developing field. This review aims to offer complete insights into the action item identification especially in the context of virtual meetings by studying various techniques, challenges, and advancements in the field of action item extraction, detection, or identification.

Traditionally, detecting action items had been viewed as a binary classification problem at the sentence level, with the goal of determining if a particular sentence contained actionable items. Prior studies (Gupta & Brenier, 2006) had used publicly available meeting datasets such as ICSI (Janin et al., 2003) and AMI (Mccowan et al., 2005) to explore a variety of machine learning approaches and feature engineering techniques. Due to the effectiveness of the fine-tuning studies of pretrained models and the renewed interest in meeting-oriented research, Yun-Nung et al. Chen, Hakkani-Tur, and He (2015) had investigated the issue of action item identification in meetings by employing Convolutional Deep Structured Semantic Model (CDSSM) to learn the latent semantics for human actions and utterances from human-machine and human-human interactions.

With the outbreak of the COVID-19 epidemic, there had been an increasing need to employ action item extraction techniques. There was a significant paradigm change as more and more companies adopted remote work policies, which increased the importance of virtual meetings as the primary means of communication. But as the number of daily meetings increased, it became necessary to extract relevant data — like important assignments — from the massive amount of information produced by these meetings. In response to the current need for action item extraction, there had been a noticeable rise in research interest in this area, indicating a major advancement in fulfilling the evolving requirements of the remote work environment of today.

A computational approach for recognizing action-related dialogue acts in multi-party meeting transcripts, such as action items, decisions, and ideas, was given by Sheshadri (Suhash Sheshadri, 2019). They used Bidirectional Encoder Representations from Transformers (BERT) for this task and presented the versatility of BERT by showing better performance of BERT based models as compared to the other state-of-the-art models. Afterwards, Sachdeva et al. Kishan Sachdeva (2021) discussed about action item detection in meetings using pretrained transformer-based models like BERT and ETC. Using the AMI meeting corpus as a baseline, they created a demonstration of how these models improved the state-of-the-art on the ICSI simulated meeting corpus. They highlighted the challenges of developing action item recognition methods that are dependable due to small data sets, noisy annotations, and sparse positive labels, and they suggested two labeling techniques: individual element classification and labeling in sequence. In contrast, Gishamer's (Gishamer, 2021) investigated automatic action item recognition in meeting transcripts by combining machine learning techniques with syntactic and semantic components and demonstrated how their strategy was effective than other cutting-edge approaches by comparing it with them. In the mean time, a dataset named CLIP (Mullenbach et al., 2021) was introduced which contained clinical action items annotated over MIMIC-III, the largest publicly available collection of real clinical notes. This dataset, curated by physicians, encompassed 718 documents, comprising 100 K sentences. They framed the task of extracting action items from these documents as multi-aspect extractive summarization, in which each component represented a different kind of action.

During the last one year, a study (Huang et al., 2023) identified sentences with actionable tasks by providing a thorough solution for the Action Item Detection Track for the General Meeting Understanding and Generation (MUG) challenge, which was held at the ICASSP 2023 Signal Processing Grand Challenge. In order to tackle this, a simple yet effective method for modeling context and speech information simultaneously was developed. The validation set's absolute F1 score increased by 0.62, indicated that the experimental results demonstrated notable improvements over baseline models. Another study (Liu, Deng, Zhang, Chen, & Wang, 2023) presented a Context-Drop approach to increase the robustness and accuracy of action item recognition in meeting transcripts. The approach used both local and global settings through contrastive learning. The first Chinese meeting corpora with manual action item annotations was created and made available by them. Although improvements in dialog summarization by LLMs offered promise for enhancing meeting experiences, these models faced challenges in processing long transcripts and customizing summaries for different user settings. An innovative meeting recap system (Asthana et al., 2023) was designed to address these problems, helping to capture important decisions, information, and action items discussed during a meeting. A study on action-item-oriented summaries was conducted very recently (Golia & Kalita, 2024), in which the action-item extraction technique was used for each meeting section sequentially. They presented an entirely novel algorithm designed to produce abstractive summaries based on the action items discussed in the meeting. These approaches represented a shift towards utilizing deep learning techniques and leveraging the rich representational power of pretrained language models in the field of action item detection. A brief summary of the above mentioned research studies has been presented in the Table 1. Table 2 is showing a comparison of existing corpora for action items.

From the review of the literature, it is interesting to note that no work has been done explicitly in the Urdu language for action item detection, despite advancements in this field. Neither has there been any effort to create datasets specifically for Urdu meetings nor has there been any research conducted on this topic in Urdu. This highlights a significant research and development gap in the Urdu language domain, especially in addressing the need for action item identification within Urdu-speaking contexts. Our research fills this gap by presenting the first corpus of action items specifically designed for Urdu meetings, along with a thorough comparative study of different deep learning and machine learning models in this context.

Table 1

Summary of the research studies for action items identification.

Research	Year	Dataset	Methodology	Language
Suhas Sheshadri (2019)	2019	AMI Meeting Corpus	LGBM ^a (Ke et al., 2017) and BERT ^b (Wang, Sun, Fu, Zhu & Tian, 2024)	English
Kishan Sachdeva (2021)	2021	AMI and ICSI Meeting Corpora	BERT and ETC ^c (Ainslie et al., 2020)	English
Flurin et al. Gishamer (2021)	2021	ICSI Meeting Corpus	SVM with Tf-idf Vectorizer	English
James et al. Mullenbach et al. (2021)	2021	CLIP Dataset from MIMIC-III	CNN and BERT-Based Baselines	English
Huang et al. (2023)	2023	AliMeeting4MUG Corpus (AMC) (Zhang et al., 2023)	Struct-BERT (Wang et al., 2019)	Mandarin
Liu et al. (2023)	2023	AMI and Chinese Meeting Corpora (AMC) ^d	Context-Drop Approach and Lightweight Model Ensemble Method	English and Chinese
Asthana et al. (2023)	2023	AMI and ICSI Meeting Corpora	DeBERTa with 12 Transformer Layers (He, Liu, Gao, & Chen, 2021) and Fine-tuned BART (Lewis et al., 2020)	English
Logan et al. Golia and Kalita (2024)	2024	AMI Meeting Corpus	BERT	English

^a Light Gradient Boosting Machine.^b Bidirectional Encoder Representations from Transformers.^c Extended Transformer Construction.^d <https://www.modelscope.cn/datasets/modelscope/Alimeeting4MUG/summary>.**Table 2**

A comparison of existing corpora for action items (Information about AMI and AMC has been taken from (Liu et al., 2023) and about ICSI has been taken from (Chen et al., 2015)).

Corpus	Year	No. of meetings	No. of utterances	No. of action items	Average no. of action items per meeting	Kappa coefficient	Language
ICSI (Janin et al., 2003)	2003	22	21 035	318	–	0.64	English
AMI (Mccowan et al., 2005)	2005	101	80 298	381	3.77	–	English
AMC (Zhang et al., 2023)	2023	424	306 846	1506	3.55	0.47	Mandarin
CLE (Ours) (Sadia, Adeeba, Shams, & Javed, 2024)	2024	240 (Work in Progress)	104 509	2098	3.98	0.78	Urdu

2.1. Disposition from the existing work

Our work represents differences from previous studies, particularly from works like (Asthana et al., 2023) and Huang et al. (2023), which are relevant to our action item detection work. Interestingly, previous research has mainly focused on languages other than Urdu, while our study is designed with Urdu-speaking contexts in mind. Furthermore, in contrast to earlier research, especially (Liu et al., 2023) which has constructed and provided Chinese corpus for action items online, our study addresses the particular language difficulties presented by Urdu and provides comprehensive, publicly available guidelines as well for the action items annotation in Urdu meetings along with the publicly available dataset. These guidelines as well as the dataset serves as a valuable resource for future research in this area in addition to facilitating our own research.

In addition, our research contributes to the field by exploring the fine-tuning of deep learning models, specifically by utilizing the strength of LLMs that is missing from all the recent works (Huang et al., 2023), Asthana et al. (2023), and Liu et al. (2023). Additionally, comprehensive comparisons between various deep learning models is carried out, including LLM-based models in the context of action items identification, where the previous studies (Huang et al., 2023) and Asthana et al. (2023) are also lacking behind.

3. Dataset

In this section, the complete process for the collection and annotation of the dataset for action items identification is explained step by step along with the explanation of developed annotation guidelines with examples. Furthermore, the main characteristics of the action items are also explicitly mentioned and explained with examples.

3.1. Data overview

The corpus consists of 240 Urdu meetings, with both meeting transcripts and audio recordings included. The meetings provide an extensive depiction of real-world settings, including a variety of areas such as administrative, technical, and natural conversations. The technical meetings in particular are entirely based on real-life scenarios that involve every stage of the software development lifecycle. Zoom, Google Meet, Microsoft Teams, and other platforms are used to record the audio meetings in order to guarantee variations in background noise, audio quality, and meeting dynamics. To support additional analysis and model training, a transcript of every meeting is kept.

3.2. Data development

The main goal of our research is to construct a well-rounded dataset that represents actual meeting scenarios, fabricated scenarios, and a diverse array of action items from different real meeting. The data has been developed by the following three ways:

3.2.1. CLE meeting corpus

The starting point of this study is the CLE Meeting Corpus, which is originally created for meeting summarization (Sadia et al., 2024). This corpus is made up of a variety of meetings from real-world scenarios, including technical, administrative, and natural discussions. ASR technology transcribed the conversations from the audio recordings of these virtually held meetings, which included four to five participants per meeting and lasted an average of thirty to forty minutes. Subsequently, the meetings are annotated in the context of action items. A total of 2098 unique action items are found by experienced annotators, who effectively dealt with the challenges involved in finding action items within the CLE Meeting Corpus through careful annotation.

3.2.2. Fabricated scenario based action items

- **Scenario Design:** A range of meeting scenarios are deliberately created that covered a number of situations, including project planning, product development, client interactions, team brainstorming, project deliverables, and much more, in order to expand the diversity of our dataset. With consideration for participant roles, goals, agenda items, and potential points of discussion, each scenario accurately represented real-life meeting settings. This ensured the generation of action items relevant to actual meeting situations.
- **Action Item Generation:** Within each scenario, a range of different types of action items are developed, such as tasks, decisions, requests, follow-ups, and more. Due dates are distributed evenly across distributions to prevent bias and are determined by a variety of durations, including short-, medium-, long-, and event-based periods. In addition, action items are written in informal Urdu, replicating real-world spoken language, and assigned to particular roles inside the scenario. In all, 600 fabricated action items have been generated.

3.2.3. Collection of action items from real meetings

- **Meeting Participation:** A number of daily, weekly, and monthly meetings have been attended covering a range of domains, including technical, administrative, and team-specific settings, with the informed consent of the participants. The diverse engagement allowed for the wide range of action items to be captured. Every discussion is well documented through extensive note-taking, which captured every action item that is discussed. The number of meetings attended ensured a sufficient volume of data (e.g., 50 daily, 20 weekly, and 10 monthly) to reach the target of 250 action items by accounting for any duplications and irrelevant references.
- **Action Item Extraction:** To correctly identify and extract valid action items from the meeting, the collected notes underwent a thorough review process that ensured their completeness, clarity and relevance to the meeting context. After extraction, sentences lacking clarity, having grammatical errors, and non-actionable statements are eliminated. Subsequently, all action items are formatted consistently, including the assignee, meeting context, and due date (if applicable). A final dataset containing roughly 250 unique and useful action items extracted from real meetings is created after curation and cleaning.

3.3. Action items annotation guidelines

Annotation, or the process of marking and classifying data, is an essential stage in every research study. Proper annotation is required in order to ensure that the data is appropriately tagged and classified according to the preset criteria. In this research, the annotation of actionable items is done systematically, following a set of guidelines. This systematic approach ensures that each and every data point is correctly categorized as either actionable or not based on pre-established guidelines and criteria. Following this careful annotation process preserves the integrity of the dataset and produces more reliable and accurate results when analyzed and reviewed later.

Characteristics of the Action Items: Before annotation, It is very crucial to understand the characteristics and attributes that distinguish phrases as action items or non-action items. Action items are certain jobs, actions, or measures that must be done in order to finish a project, solve a problem, or reach a goal. These are doable tasks that are usually assigned to people or teams to complete after meetings, conversations, or project planning sessions. Action items function as useful reminders that indicate what has to be done, by whom, and by when. They are crucial for maintaining accountability, monitoring progress, and promoting efficient cooperation and coordination within a group or company. The main components that are usually present in the actionable items, as described by Gishamer (2021), are mentioned below:

- **Description:** A description of the task that is part of the action item that needs to be completed.
- **Owner:** The individual who is responsible for executing the task. This person is often determined in the context of a group decision.
- **Time Frame:** A deadline—the time by which the task is expected to be completed.
- **Agreement:** Agreement among the other group members over the execution of the specific action item.

Component	Details
Sentence	اویس نے تفصیلی منصوبے کو سینئر مینیجمنٹ سے ڈسکس کرنے کے بعد دو ہفتوں میں تیار کرنا ہے (Awais has to prepare a detailed plan within two weeks after discussion with senior management.)
Description	تفصیلی منصوبے کو تیار کرنا ہے (Detailed plan to be prepared)
Owner	اویس (Awais)
Time Frame	دو ہفتوں (Two weeks)
Agreement	سینئر مینیجمنٹ سے ڈسکس کریں (To be discussed with senior management)
Sentence	اور آپ مجھے آج ایک رف ڈاکومنٹ بنا کے دکھائیں (And you make a rough document and show it to me today.)
Description	رف ڈاکومنٹ بنا کے دکھائیں (Show by creating a rough document)
Owner	(Not present)
Time Frame	آج (Today)
Agreement	(Not present)
Sentence	عرفان آپ کا ٹارگٹ ہے کہ آپ کی ٹیم ایک موبائل ایپ ڈویلپ کرے گی نو فروری تک (Irfan your target is that your team will develop a mobile app by February 9.)
Description	موبائل ایپ ڈویلپ کرے (Develop mobile app)
Owner	عرفان اور انکی ٹیم (Irfan and his team)
Time Frame	نو فروری (9 February)
Agreement	(Not present)
Sentence	مارچ کے آخری ہفتے تک آپ نے ویب ایپ کا فرنٹ اینڈ ریڈی کر کے دینا ہے (By the last week of March you should have the front end of the web app ready.)
Description	ویب ایپ کا فرنٹ اینڈ ریڈی کر کے دینا ہے (The front end of the web app has to be prepared)
Owner	(Not present)
Time Frame	مارچ کے آخری ہفتے (Last week of March)
Agreement	(Not present)
Sentence	فنانس اور فیسلیٹیز کی ٹیموں کے ساتھ بجٹ اور لاجسٹکس کنفرم کریں (Confirm budgets and logistics with finance and facilities teams.)
Description	بجٹ اور لاجسٹکس کنفرم کریں (Confirm budget and logistics)
Owner	(Not present)
Time Frame	(Not present)
Agreement	فنانس اور فیسلیٹیز کی ٹیموں کے ساتھ (With finance and facilities teams)

Fig. 1. Examples of the action items along with all the components identified separately.

Fig. 1 shows some of the examples of action items along with their components identified separately. The noted point here is that, it is not necessary that every action item has all the characteristics or components mentioned above. It should be noted that not every action item needs to have each of the characteristics or components listed above. It is possible that, similar to the examples, certain action items have no assignee, no agreement, or no time frame despite the fact that each action item has a description.

Guidelines: Since not every statement may be assumed an action item, it is crucial to develop understanding about what type of sentences can be considered an action item. Fig. 2 shows some examples of the sentences that can and cannot be considered as action items. The guidelines for this are as follows:

Sentence Type	Sentence
Imperative Sentence	رہورٹ مکمل کر کے جمعہ تک جمع کرائیں۔ (Complete the report and submit it by Friday.)
Conditional Sentence with Clear Actions	اگر شپمنٹ میں تاخیر ہو رہی ہے تو، ویرہاؤس مینیجر کو بتائیں اور اس کے مطابق ترسیل کے شیڈول کو اپ ڈیٹ کریں۔ (If the shipment is delayed, inform the warehouse manager and update the delivery schedule accordingly.)
Action Items with Multiple Tasks	ایجنڈا تیار کریں، پریزنٹیشن کا مواد اکٹھا کریں، اور آنے والی کانفرنس کے لیے اسپیکر کے ساتھ تعاون کریں۔ (Prepare agendas, gather presentation materials, and collaborate with speakers for upcoming conferences.)
General Instructions	چیٹ بوٹ کے لینگویج ماڈلز کو ایکسپینڈ کرنے کی ضرورت ہے (Chatbot language models need to be expanded.)
Differentiating Action Items from Discussion of Upcoming Tasks	میم کل اے ایس آر کی ٹریننگ لگانی ہے، اس کے بعد اسکی ٹیسٹنگ ہے اور دستاویز تیار کرنا ہے۔ (Mam ASR training is to be done tomorrow, followed by testing and document preparation.)

Fig. 2. Examples of sentences that can and cannot be considered as action items.

- **Imperative Sentences:** Imperative sentences are direct and command-like, and they are frequently used in action items. These sentences usually start with a verb and specify the activity or task that has to be done.
- **Conditional Sentences with Clear Actions:** Conditional sentences that specify precisely what has to be done in response to a given condition can also be considered as action items.
- **Action Items with Multiple Tasks:** An action item may occasionally consist of multiple tasks or sub-tasks. In such cases, it is crucial to clearly outline each task to be performed.
- **General Instructions (Not Action Items):** Although general instructions convey guidance or information, they do not outline a specific task or course of action. Usually, they are descriptions of situations, giving background information, or suggestions without explicitly assigning a particular action.
- **Differentiating Action Items from Discussion of Upcoming Tasks:** In general, questions and discussions about tasks that people plan to complete next week or in the future are not considered as action items. Action items are particularly defined as tasks that need immediate attention and explicit assignment, even though they may also include planning, coordinating, or communicating goals.

3.4. Annotation process

Once a comprehensive understanding of action items, their characteristics, and the types of sentences that fit the requirements for action items has been formed, the next step is to annotate or tag the action items in accordance with predetermined guidelines. While some sentences could be easy to annotate, it might be challenging to decide which sentences actually count as actionable items. It is simple to mark a statement as “yes” if it satisfies the criteria for an action item when the annotation is straightforward. However, a panel consisting of five or six individuals is called upon when there is uncertainty. This panel typically consists of three graduates and three technical specialists. Each panelist casts a vote based on their understanding of the text after reading it and discussing it with the other panelists. The final decision about the annotation is based on the majority consensus of the panel members. After each sentence is labeled, the team as a whole verifies it. When discrepancies or mistakes are discovered during the verification process, the panelists discuss the issues extensively and come to an agreement on the correct annotation. This continuous approach ensures accuracy and reliability of the annotated dataset.

3.5. Final action items corpus

After combining all the data development stages, the final action items dataset is shown in the Table 3. Three categories are used to categorize the 240 meetings in the CLE Meeting Corpus: technical, administrative, and natural. Nearly 2098 action items are developed from this large corpus. Simultaneously, 600 scenario-based fabricated action items are generated, replicating real-world meeting situations across a range of settings. In order to improve our dataset, 250 action items, that are taken from real meetings, are also added, giving relevance and authenticity the most importance. A snippet of a sample meeting having utterances of action items is shown in Fig. 3.

To demonstrate the distribution of action items in different meetings, analysis has been done which actually depicts how many meetings have action items in the defined ranges. Fig. 4 provides a visual representation of the above analysis in the form of a bar

Action Item	Meeting Transcript
No	اچھا تو ٹھیکٹ ہے پھر ان کو بلا لینا Well then, call them
Yes	م: بلکہ ماں پہلے {پرسن 1} آپ ایما کریں کہ آج آج ہی سی ویز وغیرہ لیں، جو بھی آتی ہیں Rather, first {Person 1}, you should collect the available CVs from HR by today
Yes	میں اس کو گو تھرو کر لوں گی اور {پرسن 2} کو کہہ دینا کہ ان کو ماں کل صبح کال کر دوں I will go through it and ask {Person 2} to call them tomorrow morning
No	میم میں اس کو بتا دوں گی Ma'am! I would inform her
Yes	م: مزید آپ نے فائنل فیڈ بک کے ساتھ مشاورت کے بعد مجھے اویٹیلبل بجٹ کی اپڈیٹ بھی دینی ہے Further, after consultation with the finance department, you have to give me an update of the available budget

Fig. 3. A portion of a meeting transcript along with marked action items from our CLE Meeting Corpus. This is a discussion between HR and a manager about the hiring of new candidates. Here the action item utterances are further elaborated by highlighting owner, time frame, description and agreement with blue, brown, green and yellow colors respectively.

Table 3

A summary of the stats of final action items dataset.

	Meetings	Action items	Non action items
CLE Meeting Corpus (Sadia et al., 2024)	240	2098	102411
Fabricated Action Items	–	600	–
Action Items From Real Meetings	40	250	–
Total	280	2948	102411

chart where x -axis shows the range of number of action items and y -axis shows number of meetings. It can be clearly observed that majority of the meetings have relatively lower number of action items in the range of less than or equal to 2 but a very few meetings also contain action items in the range of 15 to 16. This shows a diversity in the meetings regarding number of action items depending on the discussion being done in the meeting.

The action items in the dataset are annotated sentence by sentence using the Named Entity Recognition (NER) Text Annotator.² The annotation has been done using BIO labeling scheme in order to ensure consistency in labeling. BIO labeling scheme allows precise identification of single and multi-token entities, where the starting token of each entity is marked as “B”, inside tokens as “I”, and non-entity tokens as “O”. This procedure made it possible to tag different kinds of entities within each utterance, including owner, time frame, description, and agreement. Table 5 depicts that from all the 2948 (2098+850) action item utterances, 899 of them mentioned owners, 942 mentioned time frames, 2681 added description, and 130 mentioned agreements.

In order to check the reliability of final dataset, The inter-annotator agreement, which is determined by the kappa coefficient (Warrens, 2014), must be assessed. This statistical metric assesses the level of agreement among annotators that goes above random variation. In our case, the calculation of the Kappa value between the two annotators produced a score of 0.78, as seen in Table 6 which shows the substantial agreement between the annotators. The computed Kappa values enhance the dataset's reliability for additional analysis and application. In addition to calculating the kappa coefficient to evaluate inter-annotator agreement, additional quantitative analysis has been carried out through error analysis to further evaluate the dataset's reliability. In this analysis, key metrics such as True Positives, True Negatives, False Positives, and False Negatives have been computed, along with the False Positive Rate (FPR) and False Negative Rate (FNR) to evaluate annotation accuracy and consistency. Along with these, the classification report showing precision, recall and F1 also provides a comprehensive view of the dataset's annotation quality. Table 4 depicts the results of these metrics being computed.

² <https://tecoholic.github.io/ner-annotator/>

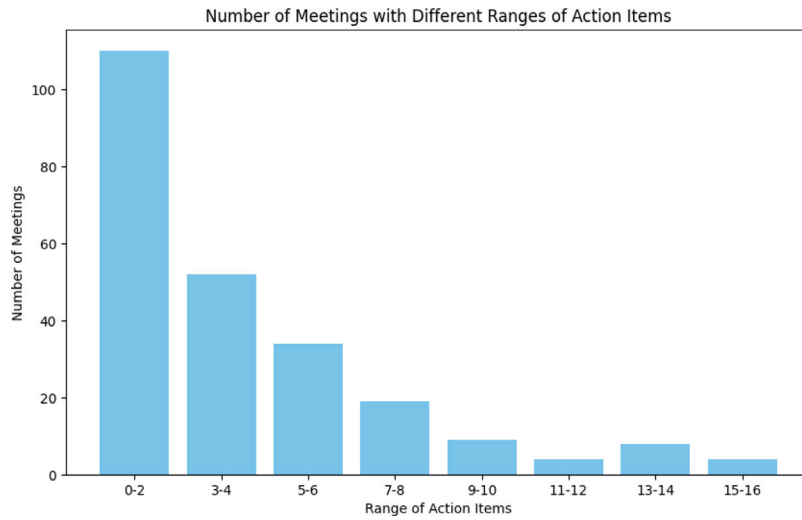


Fig. 4. Action items distribution showing number of meetings having action items in a specific range.

Table 4

Confusion matrix and classification metrics to check the reliability of action items annotation.

Metric	Non-action item	Action item	Average/Overall
Precision	0.84	0.94	0.90
Recall	0.88	0.91	0.90
F1-Score	0.86	0.92	0.90
Accuracy			0.90
Macro Average	0.89	0.90	0.89
Weighted Average	0.90	0.90	0.90
True Negatives (TN)	599		
False Positives (FP)	81		
False Negatives (FN)	118		
True Positives (TP)	1202		
False Positive Rate (FPR)	0.12		
False Negative Rate (FNR)	0.09		
Disagreement Cases	199		

Table 5

Count of each entity within total 1913 action items.

Entity Type	Count
Owner	899
Time Frame	942
Description	2681
Agreement	130

Table 6

Count for agreements and disagreements between the two annotators. Here the computed kappa co-efficient is 0.78.

	Agreement	Disagreement
Agreement	1202	118
Disagreement	81	599

The dataset demonstrates 90% agreement between annotators over 2,000 utterances, as seen in Table 4, providing a comprehensive overview on quantitative analysis of annotated dataset, which clearly demonstrates its reliability for action items detection. In addition to the False Positive Rate of 0.12 and the False Negative Rate of 0.09, the figure shows important metrics such as True Positives (1,202), True Negatives (599), False Positives (81), and False Negatives (118). As indicated by the high precision, recall, and F1-scores (all above 0.84), these values show minimal discrepancy. It can also be observed from the results that out of the total only 10% (199 utterances) of the data exhibited disagreement between the annotators highlighting areas of subjectivity in interpretation but are comparatively uncommon. All of these metrics, together with a Kappa score of 0.78, indicate the dataset's resilience and make it an effective starting point for further research in action items identification.

Sentence
تو اسی طرح سے آپ نے اپنے پلان کے اندر ٹیلیفونک کی بھی ڈالنی ہے۔ (So in the same way you have to include the telephonic in your plan.)
جی جی آؤسلی اور اس میں ہمارے لیے چیٹ بوٹ جو ہے وہ کافی امپورٹنس رکھتا ہے تو اسے ایڈ کروائیں اس والے ایڈ میں ضروری ہے۔ (Yes yes obviously and in this, chatbot is very important for us, so add it in this advertisement, its necessary.)
اور ناں آپ ڈفرنٹ برانڈز کو بھی تب تک کنسلٹ کریں ویکنڈ بھی آرہا ہے تو اگر اب پاسیبل ہے آپ جا کے پتہ کر سکتی ہیں تو جا کے کر لیں۔ (And you should consult different brands too, the weekend is also coming, so if it is possible now, you can go and find out, then go and do it.)
بجٹ جو ہے وہ ہمارے پاس دس لاکھ کا ہی ہے تو کوشش کیجیے گا کہ اس کے اندر جو ہے ناں وہ مینج ہو جائے سارا کام۔ (The budget we have is only one million, so try to manage all the work in it.)
اور احمد سے بھی بات کریں جا کر اس کو بھی بتائیں کہ مم نے یہ کہا ہے کہ ہمیں پندرہ ہیڈ فونز چاہییں۔ (And also go talk to Ahmed and tell him that Ma'am said that we need 15 headphones.)

Fig. 5. Examples of sentences that have shown disagreement among the annotators.

Fig. 5 illustrates some of the instances of annotator disagreement where general instructions are confused with action items. This confusion often emerges from the ambiguity of phrases that either indicate a specific action or remain a general statement. For example, sentences like “consult different brands” or “manage all the work in the budget”, that contain general instructions, may be read as suggestions rather than specific actions. Similar to this, statements like “tell him that we need 15 headphones” and “add it in this advertisement, it’s necessary” create ambiguity because they may be interpreted as either necessary tasks or as simple discussions about the task. These nuances led to variability in annotation, highlighting the difficulty of differentiating between general instructions and action-oriented tasks. Despite these minor annotation challenges, all of the above analyses which include the kappa coefficient, error metrics, and instances of annotator disagreement, show a high degree of data reliability overall, confirming that the dataset is suitable for precise action item identification.

The final action items dataset is a significant contribution to Urdu NLP, paving the way for more accurate and efficient action items detection in Urdu meetings. Its robust structure, extensive annotations, and diverse content allow researchers to address the challenges of action item recognition in this specific linguistic area. Its potential to assist organizations and researchers emphasizes how important it is as a resource for future explorations and advancements. The dataset is available online on github³ in csv format.

3.6. Annotator details

A total of six skilled annotators, all native Urdu speakers, are involved in the annotation process of action items within meetings. Three of them are linguistic experts, fluent in the complexities of Urdu, while the other three are technically skilled. Collectively, they achieved a comprehensive understanding of the guidelines and an accurate grasp of the task’s challenges, guaranteeing precise and consistent annotations.

3.7. Ethical consideration

Throughout the annotating process, ethical concerns are important. Prioritizing the secrecy and privacy of the data that is being examined is essential. Throughout the annotation task, the annotators remained firm in commitment to maintaining strict data protection procedures and ethical standards. Additionally, the critical step of getting informed consent from people, whose data is being used for annotation, is emphasized, which guarantees transparency and accountability throughout the annotation process.

³ https://github.com/BareeraSadia/Action_Items_Detection/tree/main/Dataset

4. Experimentation

This section provides a foundation for examining the experiments that are carried out using a variety of techniques, such as a diverse range of traditional models. Our research relies extensively on the use of LLMs. A key component of our research involves the use of LLMs, by conducting zero-shot and few-shot experiments on BLOOMZ, LLaMA, and GPT-3.5 in addition to focusing on fine-tuning BLOOMZ and LLaMA variants. The basis for training these models is the annotated dataset, which includes advanced LLMs like BLOOMZ, LLaMA as well as more conventional deep learning algorithms like CNN, LSTM, BLSTM, and transformer-based models like BERT and RoBERTa. Different evaluation metrics such as precision, recall, accuracy and F1 score are used to evaluate the performance of different models which guarantees a thorough analysis of the model's performance and its effects on the identification of action items.

4.1. Models

In this study, several traditional models have been employed that have shown effective performances in various classification and prediction tasks. These models offer a strong basis for analyzing the data while setting performance benchmarks. Support Vector Machines (SVMs) (Ruskanda, Abiwardani, Mulyawan, Syafalni, & Larasati, 2023), sometimes referred to as support vector networks in machine learning, are supervised max-margin models that evaluate data for regression and classification. Random forests (RF) (Wang, 2024), also known as random decision forests, are an ensemble learning technique that work by building a large number of decision trees during the training phase for problems like regression and classification. For classification problems where the objective is to predict the probability that an instance belongs to a specific class or not, one supervised machine learning approach that is utilized is called Logistic Regression (LR) (Kaur & Himanshu, 2023). A decision tree (Blockeel, Devos, Frénay, Nanfack, & Nijssen, 2023) constructs a tree structure resembling a flowchart, with each internal node signifying an attribute test, each branch indicating a test result, and each leaf node (terminal node) containing a class label. ConvNets, short for Convolutional Neural Networks (CNNs) (Alzubaidi et al., 2021), are specialized deep learning algorithms that can extract features on a large scale automatically. This eliminates the need for manual feature engineering, which increases efficiency. By including memory cells and gating mechanisms, Recurrent Neural Network (RNN) architectures known as Long Short-Term Memory (LSTM) (Sumathy et al., 2023) models address the issue of disappearing gradients in traditional RNNs. The Bidirectional Long Short-Term Memory (BLSTM) (Betul Polat & Cankurt, 2023), which processes sequential data in both forward and backward directions, is an adaption of the conventional LSTM design that enables it to learn longer-range dependencies in sequential data than typical LSTMs, that only process data in one direction. Utilizing unsupervised tasks like Next Sentence Prediction (NSP) and Masked Language Model (MLM), the transformer-based model BERT (Wang et al., 2024) is pretrained on massive text datasets. BERT, in contrast to other language representation models, is intended to jointly train on both left and right context in all layers in order to pre-train deep bidirectional representations from unlabeled text. By refining pretraining techniques such as eliminating NSP and training on bigger datasets with longer sequences, RoBERTa (Liu et al., 2019) improves on BERT. Lastly, to take advantage of its varied capabilities for language processing, four RoBERTa variants have been used i.e. RoBERTa-Urdu-Small (Urdu specific), XLM-RoBERTa-Base, XLM-RoBERTa-Large and Glot500-Base (remaining are multi-lingual).

LLMs (Naveed et al., 2024), trained on large text corpora, have demonstrated their ability to learn new tasks from textual instructions or a small number of examples. BLOOM is an autoregressive LLM (Scao, Fan, Akiki, Pavlick, Ilić, Hesslow, Castagné, Luccioni, Yvon, Gallé, et al., 2023) that was trained on a massive volume of text data utilizing industrial-scale computational resources to continue text from a prompt. As a result, it can produce text that is nearly identical to human-written text in 46 languages and 13 programming languages. Depending on how many parameters it has, it has several variants, including BLOOMZ-560 m, BLOOMZ-1b1, BLOOMZ-1b7, BLOOMZ-3b, and BLOOMZ-7b1. A series of autoregressive LLMs called Large Language Model Meta AI (LLaMA) (Touvron et al., 2023) was made available by Meta AI beginning in February 2023. For the initial version of LLaMA, seven, thirteen, thirty-three, and sixty-five billion parameters were used for training. LLMs of the Generative Pretrained Transformer (GPT) (Zhang, Dong, Xiao & Oyamada, 2023) type are a well-known framework for generative artificial intelligence. GPT-3 family perform admirably in the majority of NLP tasks in the general domain in zero-shot and few-shot scenarios. While other LLMs, like Google's LaMDA and PaLM and META's LLaMA, have parameter counts ranging from 50 billion to 500 billion, GPT-3.5 contains at least 175 billion parameters. An overview comprehensively showing specifications of base models for LLMs is depicted in Table 7.

The selection of these models is driven by their proven ability to handle wide range of text classification tasks in different domains. Traditional models, such as SVM, RF, LR, and Decision Trees are useful for initial analysis and comparison since they offer a stable and comprehensible baseline performance. In contrast, advanced deep learning models, such as CNN, LSTM, BLSTM, BERT and RoBERTa are included because of their exceptional capacity to recognize complex relationships and patterns in text data. In particular, CNNs work well for extracting features, but LSTMs and BLSTMs do better at modeling temporal sequences and identifying long-range dependencies in text. By utilizing bidirectional context, RoBERTa and BERT's transformer design offers an effective method for NLP that is extremely good at comprehending and categorizing human language. Adding multilingual LLMs like BLOOMZ and LLaMA alongside the GPT family makes sense, especially because our dataset is in Urdu. These models can manage the complexities of Urdu text and perform state-of-the-art tasks related to language understanding since they have been trained on large text corpora in many languages.

Table 7

An overview of the base model specifications for LLMs (Information about BLOOMZ, LLaMA2 and LLaMA3.1 has been taken from (Scao et al., 2023), Touvron et al. (2023) and Grattafiori et al. (2024) respectively).

Model	No. of parameters	No. of layers	Dimension	No. of heads	No. of tokens
BLOOMZ-560 m	559M	24	1024	16	341B
BLOOMZ-1b1	1065M	24	1536	16	341B
BLOOMZ-1b7	1722M	24	2048	16	341B
BLOOMZ-3b	3003M	30	2560	32	341B
BLOOMZ-7b1	7069M	30	4096	32	341B
LLaMA2-7b	6.7B	32	4096	32	1.0T
LLaMA3.1-8b	8B	32	4096	32	15T
GPT-3	175B	96	12 288	96	–

```

prompt = '''
You will be provided with a sentence/segment in urdu from meeting.
Your job is to identify whether the sentence/segment contains any actionable item in it or not.
Your output should be "yes" for sentence/segment that contains actionable item and "no" otherwise.

Sentence/segment : {}
Output:
'''

```

Fig. 6. Final prompt for zero-shot experiment.

4.2. Dataset

The entire dataset that is being annotated in this research contains 273 scripted meetings, 240 recorded meetings, 600 fabricated action items, and 250 action items from actual/real meetings. All of these collectively make up a dataset of about 105k utterances, of which about 3k utterances are action item utterances. This makes the skewed nature of the data clear. In order to rectify this imbalance, post-processing is initially implemented on the utterances of non-action items, which total approximately 102k. In this procedure, unique utterances are selected, and utterances having word length less than or equal to 5 are eliminated. Ultimately, a total of about 67k utterances remain, consisting of 3k action items and 64k non-action items.

For the test data, 10% of the action items (approximately 300 utterances) are selected, and an equal number of non-action items (300 utterances) are randomly chosen, resulting in a balanced test set of around 600 utterances. The remaining data is used as the training set, and 10% of the training set is used for validation. Experiments are carried out first on the complete dataset of 67k and then by using balanced training data, which consists of all the remaining action items (approximately 2.7k) and an equal number of randomly selected non-action items from the 64k utterances of non-action items. Because of the data skewness, the results of the experiments on the 67k dataset lean more towards the non-action items class. As a result, experiments are carried out on the balanced dataset and the results which are presented in the later Section 5 are actually performed on balanced data.

4.3. Implementation

The experimentation starts with the implementation of all the traditional machine learning and deep learning models mentioned in Section 4.1. These models have been trained on the balanced dataset the explanation of which is given in the above subsection. Additionally, BERT has also been fine-tuned for this classification task by using its multilingual variant available on Hugging Face.⁴ This multilingual variant of BERT supports total 102 different language in which Urdu is also included. Besides BERT, different variants of RoBERTa have also been fine-tuned including RoBERTa-Urdu-Small⁵ (specifically for Urdu), XLM-RoBERTa-Base⁶, XLM-RoBERTa-Large⁷ and Glot500-Base⁸ (multi-lingual variants of RoBERTa with Urdu support). These experiments have been conducted on Jupyter Notebook and Google Colab.

After evaluating all of the above traditional models, the LLMs including BLOOMZ-560 m, BLOOMZ-1b1, BLOOMZ-1b7, BLOOMZ-3b, BLOOMZ-7b1, LLaMA2-7b, LLaMA3.1-8b and GPT-3.5 are first evaluated using zero-shot approach in which models are asked to do the classification task of action items detection without giving any training data. For the zero-shot experiment, multiple prompts are evaluated to ensure a thorough comparison. The final prompt which has given the best results among all and is used for all the selected LLMs, is depicted in Fig. 6, while additional prompts that are tested in this case, are shown in Figs. 10, 11, and 12 added in the Appendix for reference.

⁴ <https://huggingface.co/google-bert/bert-base-multilingual-uncased>

⁵ <https://huggingface.co/urduhack/roberta-urdu-small>

⁶ <https://huggingface.co/FacebookAI/xlm-roberta-base>

⁷ <https://huggingface.co/FacebookAI/xlm-roberta-large>

⁸ <https://huggingface.co/cis-lmu/glot500-base>

```

prompt ='''
You will be provided with a sentence/segment in Urdu from meeting.
Your job is to identify whether the sentence/segment contains any actionable item in it or not.
Your output should be "yes" for sentence/segment that contains actionable item and "no" otherwise.

### A couple of examples are listed below:
1. Sentence/segment: "مریم آپ ان بلز کی کاپی کروا لینا یاد ہے۔"
   Output: yes
2. Sentence/segment: "سینسیٹیو ڈیٹا کو ہینڈل کرتے وقت ایٹھکل گائیڈ لائنز اور پرائیویسی ریگولیشنز پر پرسوں سے ہی عمل کرنا سٹارٹ کریں۔"
   Output: yes
3. Sentence/segment: "ڈیٹا کلیکشن کا سارے کام آپ نے 6 جنوری تک مکمل کرنا ہے۔ دین پھر اینوٹیشن فیز سٹارٹ کرنا ہے۔"
   Output: yes
4. Sentence/segment: "اچھا اور مسموع اللہ آپ کس کمپنی کے آرڈر کر رہے ہیں لیپ ٹاپس اور کمپیوٹرز؟"
   Output: no
5. Sentence/segment: "اولڈ امیلائز ہیں وہ دو دن وہ آئیں گے دو دن یہ آئیں گے پھر ایک دن جو ہے۔"
   Output: no
6. Sentence/segment: ""
   Output: no

### Now, classify the following sentence/segment:
Sentence/segment: {}
Output:
'''

```

Fig. 7. Final prompt for few-shot experiment.

Table 8

Fine-tuning parameters for BERT, RoBERTa-Urdu, XLM-RoBERTa, Glot500, BLOOMZ and LLaMA.

Parameter	BERT/RoBERTa-Urdu	XLM-RoBERTa/Glot500	BLOOMZ	LLaMA
Num_train_epochs	2	2	2	2
Optimizer	Adamw_torch	Adamw_torch	Adafactor	AdamW8 bit
Learning_rate	5e-05	5e-05 (1e-05 for Glot500)	2e-05	2e-05
Weight_decay	0.05	0.01	0.01	–
Per_device_train_batch_size	4	8	1	2
Per_device_eval_batch_size	4	8	8	2
Evaluation_strategy	Epoch	Epoch	No	–
Save_strategy	Steps	Epoch	No	–
Warmup_steps	0	500	0	100
Logging_steps	500	10	500	–
Logging_strategy	Steps	Steps	Steps	–
Gradient_accumulation_steps	1	1	4	1
Gradient_checkpointing	False	False	True	–
Load_best_model_at_end	False	True	False	–
Save_total_limit	None	2	None	–

Subsequently, all of these LLMs are further evaluated using few-shot approach in which models are asked to do the classification task by providing some of the examples about how this task needs to be solved. In our case, three examples are given for each of the output class (“yes” and “no”). Similar to zero-shot experiment, few-shot experiment is also evaluated on multiple prompts to ensure a comprehensive analysis. A snippet of the final complete prompt which has shown best results among all of the tested prompts is provided in Fig. 7, while additional prompts tested during this evaluation are depicted in Figs. 13, 14, and 15 in Appendix.

Finally, after the zero-shot and few-shot experiments, fine-tuning of the LLMs has been conducted on Kaggle. Due to the computational limitations of the available GPUs, only two variants of BLOOMZ i.e. BLOOMZ-560 m, BLOOMZ-1b1 and one variant of LLaMA i.e. LLaMA3.1-8b have been fine-tuned on our Urdu dataset for action items mentioned in Section 4.2. Table 8 shows all the parameters which have been setup for fine-tuning for all the models. The implementation code for fine-tuning of BERT, RoBERTa-Urdu, XLM-RoBERTa, Glot500 and BLOOMZ are made available on github.⁹ For LLaMA3.1-8b¹⁰ fine-tuning, torchtune library by Pytorch (Meta)¹¹ has been used, which provides a code free config based setup for fine tuning LLaMA based models. For this, full fine-tuning using single device has been done using config file available here¹².

All of the previous experiments focus on identifying whether a given sentence contains an action item or not. To further analyze the identified action items and to do entity-level evaluation on it, NER model has been trained which is capable of extracting specific entities from these action items, including OWNER, DESCRIPTION, TIME FRAME, and AGREEMENT. To train the NER model on

⁹ https://github.com/BareeraSadia/Action_Items_Detection¹⁰¹¹ <https://pytorch.org/torch/tune/stable/overview.html>¹² https://github.com/pytorch/torch/tune/blob/main/recipes/configs/llama3/8B_full_single_device.yaml

Table 9

Results of traditional models using precision, recall and F1 score. Models being trained on balanced dataset are evaluated on test data of around 600 utterances. Here, Class 0 (representing non-action items) belongs to the no class, Class 1 (representing action items) belongs to the yes class and the Avg. column is for macro average.

Model name	Precision			Recall			F1		
	Class 0	Class 1	Avg.	Class 0	Class 1	Avg.	Class 0	Class 1	Avg.
SVM	0.90	0.90	0.90	0.91	0.89	0.90	0.90	0.90	0.90
RF	0.84	0.95	0.90	0.96	0.81	0.89	0.90	0.88	0.89
LR	0.87	0.89	0.88	0.90	0.86	0.88	0.88	0.88	0.88
Decision Tree	0.74	0.78	0.76	0.81	0.70	0.76	0.77	0.74	0.76
CNN	0.92	0.84	0.88	0.83	0.92	0.88	0.87	0.88	0.88
LSTM	0.89	0.91	0.90	0.91	0.88	0.90	0.90	0.89	0.90
BLSTM	0.89	0.92	0.90	0.92	0.88	0.90	0.90	0.90	0.90
BERT	0.89	0.92	0.91	0.93	0.89	0.91	0.91	0.90	0.91
RoBERTa-Urdu-Small	0.92	0.91	0.92	0.91	0.92	0.92	0.92	0.92	0.92
XLM-RoBERTa-Base	0.95	0.92	0.93	0.91	0.95	0.93	0.93	0.93	0.93
XLM-RoBERTa-Large	0.98	0.89	0.93	0.88	0.98	0.93	0.93	0.93	0.93
Glott500-Base	0.94	0.93	0.93	0.93	0.94	0.93	0.93	0.93	0.93

the annotated dataset described in Section 3.5, the spacy library¹³ has been used. The trained model has been then tested on two datasets: (1) the action items from the test dataset with ground truth labels, and (2) the action items predicted by the best-performing model, ur_BLOOMZ-1b1. The training and testing code for the NER model is made available on github¹⁴ for researchers.

5. Results and discussion

Metrics for evaluation are essential for evaluating the performance of different models. In this research, several commonly used evaluation metrics, including precision, recall, and F1 score. With the use of these evaluation metrics, the performance of our models can be better understood and the effectiveness in addressing our research problem can be observed. The explanation of these evaluation metrics are available in the points below:

- **Precision** measures the ratio of correctly predicted positive class to the total predicted positives. The formula for the calculation of precision is provided in Eq. (1). Here “TP” means true positive and “FP” means false positive.

$$Precision = \frac{TP}{TP + FP} \quad (1)$$

- **Recall**, which is also known as true positive rate or sensitivity, measures the ratio of correctly predicted positive class to all the actual positives. Its formula is provided in Eq. (2). Here “TP” means true positive and “FN” means false negative.

$$Precision = \frac{TP}{TP + FN} \quad (2)$$

- **F1 score** is the harmonic mean of precision and recall, providing a balanced measure between the two. F1 score is calculated using Eq. (3).

$$Precision = 2 \times \frac{Precision \cdot Recall}{Precision + Recall} \quad (3)$$

Table 9 shows results of traditional models in the form of precision, recall and F1 score for action items (Class 1), non-action items (Class 0) and macro average of all the three measures. Results of LLMs for all the three settings i.e. zero-shot, few-shot and fine-tuned are depicted in Table 10 in the form of precision, recall and F1 score for action items (Class 1), non-action items (Class 0) and macro average of all the three measures.

5.1. Discussion on results of traditional models

Our experiments with traditional models demonstrate that the SVM model produced an average precision of 0.90, with the same precision for non-action items (Class 0) and action items (Class 1) i.e. 0.90. With an average recall and F1 score of 0.90, the results are more evenly distributed. RF also demonstrated a balanced performance with an average precision of 0.90, an average recall of 0.89, and an average F1 score of 0.89, demonstrating its robustness in both classifications. Among the models LR and Decision Tree, LR maintained a higher precision (0.89) and recall (0.86) for action items (Class 1) as compared to Decision Tree (0.78) and (0.70) respectively, resulting in a better average F1 score for LR (0.88) in comparison to Decision Tree (0.76).

In general, deep learning models, including CNN, LSTM, and BLSTM, CNN’s average F1 score is 0.88, while the LSTM and BLSTM models performed even better, averaging 0.90 for both. BERT produced an average F1 score of 0.91 whereas RoBERTa-Urdu, XLM-RoBERTa and Glott500 showed an average F1 score of 0.92, 0.93 and 0.93 respectively depicting that their performance is even

¹³ <https://pypi.org/project/spacy/>

¹⁴ https://github.com/BareeraSadia/Action_Items_Detection

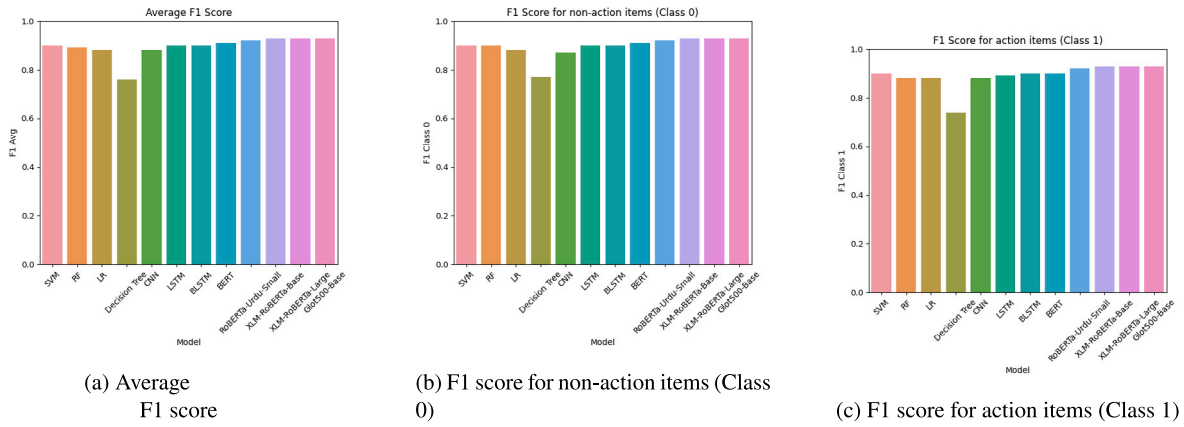


Fig. 8. Results for traditional models demonstrated in the form of F1 scores of all the models for non-action items (Class 0), action items (Class 1) and average of both.

Table 10

Results of LLMs and GPT-3.5 using precision, recall, F1 score and accuracy. Models are evaluated on three types of experiments: zero-shot, few-shot and fine-tuned. These results have been computed on test data of around 600 utterances with models being trained on balanced dataset. Here, Class 0 (representing non-action items) belongs to the no class, Class 1 (representing action items) belongs to the yes class and the Avg. column is for macro average.

Experiment	Model name	Precision			Recall			F1		
		Class 0	Class 1	Avg.	Class 0	Class 1	Avg.	Class 0	Class 1	Avg.
Zero-shot	BLOOMZ-560 m	0.47	0.48	0.47	0.21	0.75	0.48	0.29	0.59	0.44
	BLOOMZ-1b1	0.52	0.50	0.51	0.29	0.73	0.51	0.37	0.59	0.48
	BLOOMZ-1b7	0.57	0.54	0.56	0.49	0.62	0.55	0.52	0.58	0.55
	BLOOMZ-3b	0.54	0.51	0.53	0.36	0.68	0.52	0.43	0.59	0.51
	BLOOMZ-7b1	0.53	0.51	0.52	0.33	0.70	0.52	0.41	0.59	0.50
	LLaMA2-7b	0.00	0.49	0.25	0.00	1.00	0.50	0.00	0.66	0.33
	LLaMA3.1-8b	0.94	0.60	0.77	0.37	0.98	0.67	0.53	0.74	0.64
Few-shot	GPT-3.5	0.86	0.61	0.74	0.43	0.93	0.68	0.57	0.74	0.65
	BLOOMZ-560 m	0.54	0.49	0.52	0.18	0.84	0.51	0.26	0.62	0.44
	BLOOMZ-1b1	0.51	0.50	0.50	0.26	0.75	0.50	0.34	0.60	0.47
	BLOOMZ-1b7	0.57	0.53	0.55	0.41	0.68	0.55	0.48	0.59	0.54
	BLOOMZ-3b	0.50	0.49	0.49	0.53	0.46	0.49	0.51	0.47	0.49
	BLOOMZ-7b1	0.61	0.57	0.59	0.50	0.67	0.59	0.55	0.61	0.58
	LLaMA2-7b	0.63	0.50	0.57	0.04	0.98	0.51	0.08	0.66	0.37
Fine-tuned	LLaMA3.1-8b	0.92	0.60	0.76	0.37	0.97	0.67	0.53	0.74	0.64
	GPT-3.5	0.98	0.61	0.80	0.37	0.99	0.68	0.54	0.75	0.65
	ur_BLOOMZ-560 m	0.95	0.91	0.93	0.90	0.96	0.93	0.93	0.93	0.93
	ur_BLOOMZ-1b1	0.96	0.92	0.94	0.92	0.96	0.94	0.94	0.94	0.94
	ur_LLaMA3.1-8b	0.83	0.65	0.74	0.51	0.89	0.70	0.63	0.75	0.69

more better than others. Among all of these traditional models, RoBERTa variants have shown best performance in this task of identification of action items, especially XLM-RoBERTa and Glot500 with an average F1 score of 0.93. Fig. 8 provides a visual representation of F1 scores for non-action items (Class 0), action items (Class 1) and average of both.

5.2. Discussion on results of LLMs

Experiments using zero-shot, few-shot, and fine-tuned models demonstrate clear differences in performance across various settings. With no task-specific training, the GPT-3.5 model demonstrated a reasonably strong performance for zero-shot experiments, with the highest average F1 score of 0.65. The performance of the BLOOMZ models varied, with BLOOMZ-1b7 outperforming its counterparts in terms of achieving a better balance between precision and recall. In particular, BLOOMZ-1b7 performed better consistently in the zero-shot context with an average F1 score of 0.55. Among the LLaMA variants, LLaMA3.1-8b has shown a better performance with an average F1 score of 0.64. LLaMA3.1-8b has shown best precision of 0.77 among all the LLMs for zero-shot experiment.

For the majority of models, the few-shot setting produced better results. With an average F1 score of 0.65, GPT-3.5 once more demonstrated better performance than others. With an average F1 score of 0.58, BLOOMZ-7b1 likewise demonstrated a significant improvement, suggesting that few-shot learning improved the model's comprehension of the problem. LLaMA3.1-8b, with an average

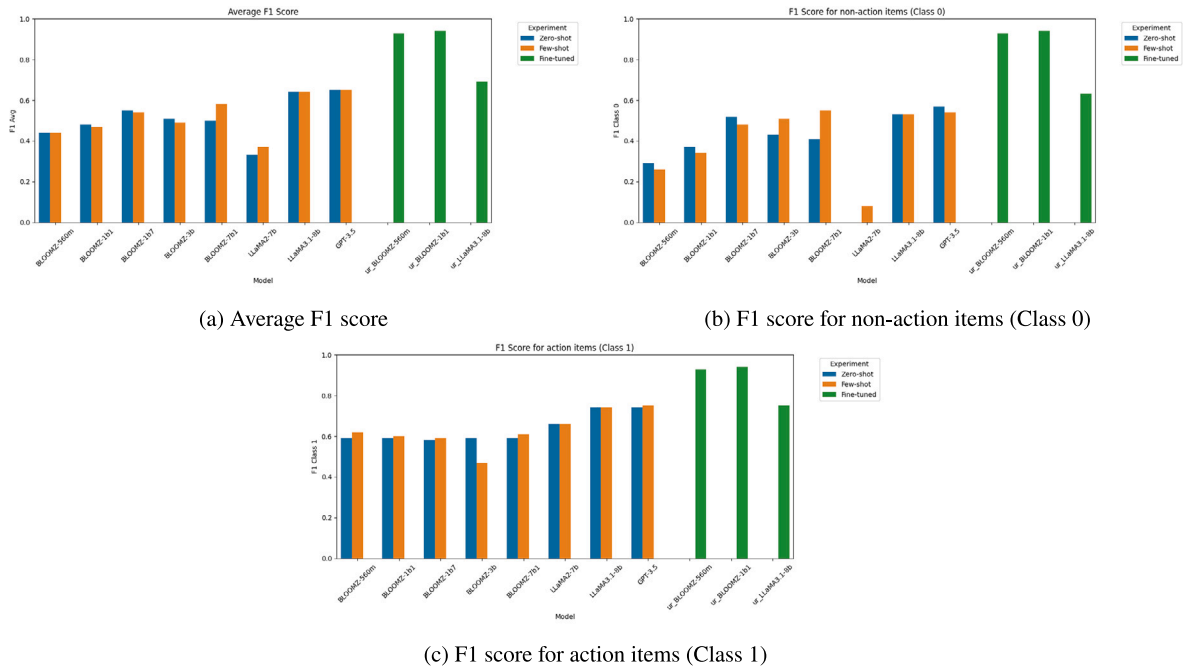


Fig. 9. Results for LLMs demonstrated in the form of F1 scores of all the models and experiments (zero-shot, few-shot and fine-tuned) for non-action items (Class 0), action items (Class 1) and average of both.

Table 11

NER performance on test data (ground truth action items).

Label	Precision	Recall	F1 score
OWNER	0.8875	0.9103	0.8987
DESCRIPTION	0.4529	0.2750	0.3422
TIME FRAME	0.5181	0.5181	0.5181
AGREEMENT	0.7500	0.1304	0.2222
Overall	0.5757	0.4181	0.4844

Table 12

NER performance on predicted action items from best performing model i.e. ur_BLOOMZ-1b1.

Label	Precision	Recall	F1 score
OWNER	0.8987	0.9103	0.9045
DESCRIPTION	0.4578	0.2764	0.3447
TIME FRAME	0.5250	0.5122	0.5185
AGREEMENT	0.7500	0.1304	0.2222
Overall	0.5836	0.4192	0.4879

F1 score of 0.64, has shown a consistent behavior for few-shot experiment. Even while LLaMA2-7b demonstrated a better F1 score for action items (Class 1) in comparison to all variants of BLOOMZ, its limited recall for non-action items (Class 0) in both zero-shot and few-shot scenarios resulted in a low average F1 score.

The models' performance is significantly improved by fine-tuning them for the particular task. The fine-tuned models ur_BLOOMZ-560 m, ur_BLOOMZ-1b1 and ur_LLaMA3.1-8b have average F1 scores of 0.93, 0.94 and 0.69, respectively. Among these three, ur_BLOOMZ-1b1 has outperformed all the other models having highest average precision, recall and F1 score of 0.94 each. Fig. 9 depicts a visual demonstration of all the model results performed on zero-shot, few-shot and fine-tuned setup. It can be clearly observed that fine-tuning has very much improved the performance of LLMs in the context of detection of action items.

5.3. Discussion on results of NER

The performance of the NER model was evaluated on two scenarios: using ground truth action items from test data and using predicted action items from the ur_BLOOMZ-1b1 model. The results, summarized in Tables 11 and 12, provide precision, recall and F1 score of the NER model being evaluated on prediction of four entities i.e. OWNER, DESCRIPTION, TIME FRAME, AGREEMENT.

5.3.1. Using ground truth action items from test data

The NER model's results are shown in Table 11 which demonstrates that the model overall achieved precision, recall, and F1 score of 0.5757, 0.4181, and 0.4844, respectively. The model demonstrated strong performance for the OWNER label (F1: 0.8987) and moderate performance for TIME FRAME (F1: 0.5181). However, its performance for DESCRIPTION (F1: 0.3422) and AGREEMENT (F1: 0.2222) was weaker, indicating challenges in identifying these categories consistently.

5.3.2. Using predicted action items from ur_BLOOMZ-1b1 model

The NER trained model was also tested on action items predicted by the best performing model which in our case is ur_BLOOMZ-1b1 model. This model achieved an average F1 score of 0.94 on action item identification. The results of NER model on these predicted action items are depicted in Table 12. It can be observed that despite the input being less structured, the NER model achieved slightly improved metrics with an overall precision of 0.5836, recall of 0.4192, and F1 score of 0.4879. Performance for OWNER (F1: 0.9045) and TIME FRAME (F1: 0.5185) remained consistent, showing the robustness of the NER model.

6. Conclusion

In the context of action item detection in Urdu meetings, this research brings numerous significant advances to the fields of NLP. First, not only comprehensive annotation guidelines have been created but also an extensive dataset has been developed which is remarkably the first ever dataset of action items for Urdu meetings. Secondly our research not only compares several traditional models' performance but also evaluates LLMs using zero-shot and few-shot setup, with BLOOMZ and LLaMA being specifically fine-tuned to enhance their performance in recognizing action items in Urdu meetings.

Our findings of the research demonstrate that fine-tuned LLM ur_BLOOMZ-1b1 has outperformed all the traditional models and pretrained LLMs by achieving highest f1 score for both the classes which reveals that its ability to handle multilingual text classification tasks can be enhanced by fine-tuning. For researchers and practitioners, our study not only contributes to the understanding of action item identification in Urdu meetings but also provides a robust framework for future studies and implementations in similar multilingual environments.

7. Limitations

Acknowledging certain limitations is crucial while introducing the first-ever dataset of Urdu action items. First of all, the data is domain-specific, having been composed from meetings mostly in the administrative domains such as hiring, procurement, admin affairs and finance. The selection of these specific domains is clearly justified in Sadia et al. (2024). Secondly, the data exhibits skewness due to a noticeable imbalance, with a higher proportion of non-action items than action items. This imbalance makes it difficult to create models that effectively identify and compile action items.

Despite these drawbacks, this annotated CLE Meeting Corpus for the action items detection is a powerful source not only for the development of action items detection systems in Urdu code-mixed language as well as for further research in this area. Even though the current focus is on specific domains, the corpus provides a solid foundation for research in this domain and can be expanded further for more improvements. It is important for researchers to be aware of these limitations while evaluating the corpus's suitability for their particular use cases.

8. Future work

Regarding future work, the action items annotated CLE Meeting Corpus will see two major areas of improvement. At first, the dataset will be enlarged to include a wider variety of meeting scenarios and domains. More meetings from different domains will be tagged to make the corpus more generic and useful. By adding additional action items from a wider range of real-world meetings, the dataset will become more comprehensive and advantageous for training action item identification algorithms.

Furthermore, a major task in the future will involve rephrasing action items to further enhance their context and clarity. This includes improving the phrasing of action items so that they are more clearly understood and do not appear as random lines in the conversation. In this way, action items will be effectively communicated by making them clearer and more informative. This enhancement will support the development of systems that are better able to identify and display action items. All of these improvements will make the CLE Meeting Corpus even more useful and applicable in the field of action items detection.

CRediT authorship contribution statement

Bareera Sadia: Writing – original draft, Software, Methodology, Investigation. **Farah Adeeba:** Writing – review & editing, Supervision, Funding acquisition, Conceptualization. **Sana Shams:** Project administration, Data curation. **Sarmad Hussain:** Supervision, Resources.

Appendix

See Figs. 10–15.

```
prompt ='''
You will be provided with a sentence/segment. You have to identify whether the sentence/segment contains any actionable item in it or not.
Your output should be "yes" for sentence/segment that contains actionable item and "no" otherwise.
Sentence/segment : {}
Output:
'''
```

Fig. 10. Prompt1 for zero-shot experiment.

```
prompt ='''
You will be provided with a sentence/segment. You have to identify whether the sentence/segment contains any actionable item in it or not.
Sentence/segment : {}
Output:
'''
```

Fig. 11. Prompt2 for zero-shot experiment.

```
prompt ='''
Your job is to identify whether the given sentence contains any actionable item in it or not.
Your output should be "yes" for sentence that contains actionable item and "no" otherwise.
Sentence/segment : {}
Output:
'''
```

Fig. 12. Prompt3 for zero-shot experiment.

```
prompt ='''
You will be provided with a sentence/segment.
You have to identify whether the sentence/segment contains any actionable item in it or not.
Your output should be "yes" for sentence/segment that contains actionable item and "no" otherwise.

### Examples:
1. Sentence/segment: "مریم آپ ان بلز کی کاپی کروا لینا یاد ہے۔"
   Output: yes
2. Sentence/segment: "سینسٹیو ڈیٹا کو ہینڈل کرتے وقت اینتھکل گائیڈ لائنز اور پرائویسی ریگولیشنز پر پرسوں سے ہی عمل کرنا سٹارٹ کریں۔"
   Output: yes
3. Sentence/segment: "ڈیٹا کلیکشن کا سارے کام آپ نے 6 جنوری تک مکمل کرنا ہے دین پھر اینوٹیشن فیز سٹارٹ کرنا ہے۔"
   Output: yes
4. Sentence/segment: "اچھا اور مسمیع اللہ آپ کس کمپنی کے آرڈر کر رہے ہیں لیپ ٹاپس اور کمپیوٹرز"
   Output: no
5. Sentence/segment: "میلانز ہیں وہ دو دن وہ آئیں گے پھر ایک دن جو ہے تھوڑا سا مکس اپ کر کے تو اس طرح سے مینج کر لیتے ہیں"
   Output: no
6. Sentence/segment: ""
   Output: no
Sentence/segment: {}
Output:
'''
```

Fig. 13. Prompt1 for few-shot experiment.

```
prompt ='''
You will be provided with a sentence/segment.
You have to identify whether the sentence/segment contains any actionable item in it or not.

### Examples:
1. Sentence/segment: "مریم آپ ان بلز کی کاپی کروا لینا یاد ہے۔"
   Output: yes
2. Sentence/segment: "سینسٹیو ڈیٹا کو ہینڈل کرتے وقت اینتھکل گائیڈ لائنز اور پرائویسی ریگولیشنز پر پرسوں سے ہی عمل کرنا سٹارٹ کریں۔"
   Output: yes
3. Sentence/segment: "ڈیٹا کلیکشن کا سارے کام آپ نے 6 جنوری تک مکمل کرنا ہے دین پھر اینوٹیشن فیز سٹارٹ کرنا ہے۔"
   Output: yes
4. Sentence/segment: "اچھا اور مسمیع اللہ آپ کس کمپنی کے آرڈر کر رہے ہیں لیپ ٹاپس اور کمپیوٹرز"
   Output: no
5. Sentence/segment: "میلانز ہیں وہ دو دن وہ آئیں گے پھر ایک دن جو ہے تھوڑا سا مکس اپ کر کے تو اس طرح سے مینج کر لیتے ہیں"
   Output: no
6. Sentence/segment: ""
   Output: no
Sentence/segment: {}
Output:
'''
```

Fig. 14. Prompt2 for few-shot experiment.

```

prompt ='''
Your job is to identify whether the given sentence contains any actionable item in it or not.
Your output should be "yes" for sentence that contains actionable item and "no" otherwise.

### Examples:
1. Sentence/segment: "مریم آپ ان بلز کی کاپی کروا لینا یاد ہے۔"
   Output: yes
2. Sentence/segment: "سینسٹرو ڈیٹا کو جنڈل کرتے وقت ایٹھکل گائیڈ لائنز اور برائویس ریگولیشنز پر برسوں سے جی عمل کرنا سٹارٹ کریں۔"
   Output: yes
3. Sentence/segment: "ڈیٹا کلیکشن کا سارے کام 6 جنوری تک مکمل کرنا ہے۔ دین پھر اینوٹیشن فیز سٹارٹ کرنا ہے۔"
   Output: yes
4. Sentence/segment: "اچھا اور مسمیہ اللہ آپ کس کمپنی کے آرڈر کر رہے ہیں لیپ ٹاپس اور کمپیوٹرز"
   Output: no
5. Sentence/segment: "میلانز ہیں وہ دو دن وہ آئیں گے دو دن یہ آئیں گے پھر ایک دن جو ہے تھوڑا سا مکس اپ کر کے تو اس طرح سے مینج کر لیتے ہیں۔"
   Output: no
6. Sentence/segment: ""
   Output: no
Sentence/segment: {}
Output:
'''

```

Fig. 15. Prompt3 for few-shot experiment.

Data availability

Data will be made available on request.

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